Arnav Yayavaram

arnav.yayavaram@gmail.com | (412) 689-0495
Work Authorization: US Citizen
ayayavar@andrew.cmu.edu

Education

Dec 2026 Pittsburgh, PA	**Carnegie Mellon University** Master of Science in NLP/ML, Language Technologies Institute, School of Computer Science *Current Coursework: Advanced Natural Language Processing, Generative AI, Machine Learning*
July 2025 Pilani, India	**Birla Institute of Technology and Science, Pilani** B.E. in Computer Science (**CGPA: 9.91/10**, Institute **Silver** Medalist)

Publications [All First/Co-First Author]

ICCV'25, *ACL	**CAIRE: Cultural Attribution of Images by Retrieval-Augmented Evaluation**. CEGIS @ ICCV'25, ACL Rolling Review (Accept, to be submitted to *ACL 2026) | Paper
LREC-COLING'24	**BERT-based Idiom Identification using Language Translation and Word Cohesion**. Multiword Expressions and Universal Dependencies @ LREC-COLING'24 | Paper
IEEE IS'24	**Interpretable Feature Optimization for Sadness Recognition in Speech Emotion Analysis**. IEEE 12th International Conference on Intelligent Systems (IS) | Paper

Experience

Carnegie Mellon University, Machine Learning Department — Pittsburgh, PA
Graduate Student Researcher — Aug 2025 – Present

- Developing a multimodal benchmark for agentic game development with annotated tasks and automated evaluation pipelines.
- Automating task and test generation, evaluating agentic LLM baselines for tool-use and fine-grained code generation.

Carnegie Mellon University, Language Technologies Institute — Pittsburgh, PA
Research Intern (Undergraduate Thesis), NeuLab | Advisor: Prof. Graham Neubig | Code — May 2024 – Mar 2025

- Developed a novel metric to quantify cultural relevance of real and generated images, and built an efficient large-scale (6 million entities) text-disambiguation image retrieval system using FAISS, surpassing SOTA LVLMs on the FOCI benchmark.
- Augmented LLMs with retrieved cultural context and Chain-of-thought prompting to compute relevance across cultural proxies, achieving **+28% F1** on a challenging hand-curated validation set. Achieved Pearson $r > 0.65$ vs human annotations on a dataset comprising universal concepts. Accepted @ **ICCV-W** & **ARR Accept** (recommended for ***ACL**).

Nanyang Technological University — Singapore
Research Intern, SpeechLab | Advisor: Prof. Chng Eng Siong | Code — Mar 2024 – Sep 2024

- Fine-tuned LLaMA-3.1-8B with LoRA on the DAIC-WOZ dataset for text-based depression detection, achieving a **+7.1%** F1 improvement over prior work. Designed a PHQ-8–guided prompting strategy, enhancing both accuracy & interpretability.

Amazon, Applied Science — Bangalore, India
Summer Intern | Advisor: Jitenkumar Rana | Code — May 2023 – Aug 2023

- Developed and validated a product brand and model knowledge base using BERT-based NER, applied data augmentation with GPT and Falcon 7B, and created regression-based metrics to detect anomalies in shipping costs.

BITS Pilani — India
Research Assistant | Engaged in **4** Research Projects in Machine Learning–based Systems — July 2023 – May 2025

- **BERT-based Idiom Detection:** Designed custom loss functions to improve token-level idiom recognition. | Code
- **Interpretable SER:** Metaheuristic feature selection for emotion detection; SOTA F1 across 4 popular datasets | Code
- **Malware Detection:** GNN/Sequence models for multi-class classification on obfuscated malware datasets | Code
- **In-Context-Learning with Information Retrieval:** Critically evaluated the methodology of the ECIR Best Paper, identifying flaws and proposing corrections, achieving improved performance on downstream NLP classification tasks.

Projects

⋆ **Pittsburgh Q&A RAG System** — Sep 2025 – Oct 2025
Built a Retrieval-Augmented Generation (RAG) pipeline using Qwen2.5 for answering Pittsburgh-related questions, integrating hybrid retrieval (dense + sparse) methods. Scraped and preprocessed source data for high-quality, domain-specific retrieval.

⋆ **Basic PASCAL Compiler | Code** — Jan 2024 – May 2024
Implemented a simplified Pascal compiler with LEX/YACC: lexer, parser, semantic checks and intermediate code generation.

Skills

Programming & OS: Python, C/C++, Java, SQL, Linux, High Performance Computing Clusters (HPC)
Libraries and Frameworks: PyTorch, TensorFlow, Numpy, Pandas, Scikit-Learn, HuggingFace, Matplotlib, spaCy